

Maximilian Lipski
Graphics Programmer

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Education

Technical University Dresden

Master of Science, Computer Science
Specialization: Computer Graphics
Expected Graduation: 01/2027

Dresden, DE

Heinrich-Heine University

Bachelor of Science, Computer Science
Thesis: Development of a Skew-Heap Data Structure for Persistent Memory

Düsseldorf, DE
10/2020 - 09/2024

Experience

Student Research Assistant

TU Dresden - [ScaDS.AI](#)

Dresden, DE
11/2025 - 04/2026

- Researched machine learning approaches for accelerating geometric computations in computer graphics.
- Developed and trained neural networks for geometric approximation problems using CUDA-enabled machine learning workflows.
- Investigated neural representations of Hermite spline tubes as signed distance functions.
- Designed and evaluated experiments to analyze the accuracy and performance of learned geometric approximations.
- Achieved promising results for fixed primitives while identifying parameterization as a major challenge or generalization

Projects

Rodan - Real-Time Rendering Engine

- Designed and implemented a real-time rendering engine built on top of a custom Rendering-Hardware Interface (RHI) with a Vulkan backend.
- Achieves around 400 FPS while rendering the Sponza scene at 2560x1600 on an RTX 5070.
- Implemented physically based rendering (PBR) pipeline with image-based lighting (IBL).
- Implemented an HDR tonemapping pipeline.
- Implemented glTF 2.0 asset loading with support for KHR_materials_transmission and KHR_materials_volume.
- Integrated Dear ImGui for in-engine tooling and debugging.
- Debugged and profiled rendering workloads using NVIDIA Nsight Graphics and RenderDoc

Velos - Rendering Hardware Interface

- Designed and implemented a backend-independent RHI with a Vulkan backend.
- Implemented abstractions for pipelines, descriptor sets, textures, samplers, buffers, and command lists.
- Built automatic SPIR-V shader reflection for descriptor set and pipeline layout generation.
- Architected the system to support future graphics backends such as Direct3D 12.

Skills & Interests

Programming: C++23, C, GLSL

Graphics: Vulkan, OpenGL, CUDA

Language: German (Native), Polish (Native), English (C1)

Tools: RenderDoc, NVIDIA Nsight Graphics, Git, CMake, Premake, Unreal Engine 5, Linux, Windows